

Modeling Positivity in the Workplace

A company wishes to study the work-life balance of its employees. To do so, they have collected data from 248 employees. You are in charge of analyzing those data and to make a written report on your conclusions, as well as of making recommendations if appropriate.

1. The data are available in an Excel file named « Base_v6_248_HD.xlsx »

The variables of this data set are the following:

- a. Individu: identification number of the respondent (a number)
- b. Sex: the gender of the respondent
 - i. The value 1 indicates a male
 - ii. The value 2 indicates a female
- c. Situfam: the family status of the respondent
 - i. 1 : Single
 - ii. 2 : Separate / Divorced
 - iii. 3 : Couple
- d. Niveaueude: The educational level of the respondent
 - i. 1: Has not completed high school degree
 - ii. 2: Has completed only a high school degree
 - iii. 3: Has completed up to 2 years after high school degree
 - iv. 4: Has completed up to 3 years after high school degree
 - v. 5: Has completed up to 4 years after high school degree
 - vi. 6: Has completed 5 years after high school degree or more
- e. Age: age of the respondent
- f. MH_Prof: the flourishing state of the respondent in an occupational context
 - i. Languishing (value = 1)
 - ii. Moderately Mentally Healthy (value = 2)
 - iii. Flourishing (value = 3)
- g. MH_Priv: the flourishing state of the respondent in a private context
 - i. Languishing (value = 1)
 - ii. Moderately Mentally Healthy (value = 2)
 - iii. Flourishing (value = 3)
- h. PsurN: the positivity ratio value, i.e., the ratio between the average positive emotions, over the average negative emotions
 - i. If the ratio is lesser than 1, the individual is diagnosed as depressed
 - ii. If the ratio is between 1 and 2, the individual is languishing
 - iii. If the ratio is between 2 and 3, the individual is moderately emotionally healthy
 - iv. If the ratio is greater than 3, the individual is emotionally flourishing
- i. MHCB_total: the total score of the respondent's mental flourishing in a professional context
- j. MHCC_total: the total score of the respondent's mental flourishing in a private context
- k. Flow_total: the Flow score of the respondent, measured as performance, mastery and focus

Problem

Positivity is considered to be a powerful flourishing indicator. People are considered flourishing when their PsurN is equal to three or greater.

Your main goal is to predict human positivity in the work environment. To accomplish this, you will first need to explore your data.

Step 0. Data preparation

Create a binary Positivity variable indicating flourishing vs non flourishing.

Take a moment to get a good understanding of your dataset. How many categorical variables do you have? Do you need some recoding before moving on to the analyses?

In Social Science, it is common to consider that ordinal variables with 5 or more categories may be considered as quantitative, especially if they measure something... Would this be applicable to your case?

Step 1. Data exploration

Provide a brief descriptive statistics of each your variables according to their types. Do you notice anything worthy of interest?

Provide a bivariate descriptive statistics of your variables. Which ones do you find relevant to explore issues related to prediction of positivity?

- Hints: Are there variables significantly linked to positivity? Are there differences in flourishing score between groups of positivity?

What variables would you expect to significantly predict positivity?

Step 2. Modeling

Build an appropriate linear model to predict positivity.

Select the best possible model based on the appropriate fit indicators (if in doubt, give priority to Schwarz BIC or SBC).

Are quadratic, interactions terms or any other transformation conducive to a better model?

Estimate the performance of your model while accounting for possible imbalance in dataset?

Write a Human Resource report destined to the leadership of the company with the relevant arguments, figures, and references.